ACADEMICS MODULE (CORE ENGINE)

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 • March 2026
Confidential • Internal Documentation

Audience: School Administrators • Teachers • IT Team • Students • Parents
Related Modules: Student Management • Attendance • Exams & Assessments • Behavior • Communication • Parent Portal

---

1. MODULE OVERVIEW

1.1 Purpose

The Academics Module manages all aspects of curriculum delivery, student performance tracking, and academic reporting. It provides a structured environment for:

- Curriculum and subject management
- Class scheduling and teacher assignment
- Continuous assessment and grading
- Report card generation
- Homework and assignment workflows
- Academic analytics and insights
- Parent and student engagement

---

2. CORE OBJECTIVES

1. Enable structured curriculum and class management
2. Track student performance across subjects and terms
3. Automate grading, report cards, and progress tracking
4. Provide real-time analytics and actionable insights
5. Link academics to attendance and behavior for holistic tracking
6. Ensure secure, role-based access for staff, students, and parents
7. Support mobile-first interaction for teachers and students

---

3. REAL-WORLD SYSTEM FLOW

3.1 Lifecycle Flow

Course Created → Subject Assigned → Student Enrolled → Assessment Entered → Grades Calculated → Report Card Generated → Performance Monitored → Parent/Student Notified

3.2 Daily Operational Flow

Teacher creates/updates assignments or assessments
↓
Students submit assignments (optional online)
↓
Teacher records grades or comments
↓
System calculates subject-level averages and trends
↓
At-risk students flagged automatically
↓
Reports sent to parents or students (if configured)

---

4. MODULE ARCHITECTURE

```
Academics Module
│
├── Curriculum & Subject Management
├── Class & Teacher Management
├── Gradebook & Assessment Tracking
├── Report Cards & Progress Reports
├── Homework & Assignments Management
├── Academic Calendar
├── Analytics & Performance Dashboards
├── Alerts & Notifications
├── Parent & Student Portal Integration
├── Data Security & Role-Based Access
```

---

5. CURRICULUM & SUBJECT MANAGEMENT

5.1 Features

- Define subjects per grade/class
- Link subjects to teachers
- Curriculum mapping (CBC, 8-4-4, custom)

- Subject weighting for grade calculations

5.2 Data Model

Subject

- subject_id
- name
- grade_level
- teacher_id
- curriculum_type
- weight

---

6. CLASS & TEACHER MANAGEMENT

6.1 Features

- Assign teachers to classes and subjects
- Create class schedules and timetables
- Track teaching hours and load
- Manage multi-section classes

6.2 Data Model

Class

- class_id
- grade_level
- section
- teacher_ids
- student_ids
- schedule

---

7. GRADEBOOK & ASSESSMENT TRACKING

7.1 Features

- Record continuous assessments, exams, and assignments

- Auto-calculate averages, totals, and term grades
- Flag at-risk students (failing 2+ subjects)
- Teacher comments integration

7.2 Real-World Example

Math: 85 (A)
English: 72 (B)
Science: 90 (A-)
Trend: Improving

7.3 Data Model

Assessment

- assessment_id
- student_id
- subject_id
- type (exam/assignment/CA)
- score
- max_score
- term
- comments

---

8. REPORT CARDS & PROGRESS REPORTS

8.1 Features

- Generate term or annual report cards
- PDF export and printing
- Parent portal access
- Automatic inclusion of grades, comments, and trends

8.2 Automation

- Pull grades from gradebook automatically
- Include attendance and behavior summary
- Notify parents when reports are available

---

9. HOMEWORK & ASSIGNMENTS

9.1 Features

- Assignment creation and distribution
- Submission tracking
- Teacher grading and feedback
- Deadline reminders and alerts

9.2 Data Model

Assignment

- assignment_id
- title
- description
- class_id
- subject_id
- due_date
- submission_status
- grade
- teacher_comments

---

10. ACADEMIC CALENDAR

10.1 Features

- Define term dates, holidays, and exam periods
- Sync with class schedules
- Trigger automated reminders for assignments, exams, and assessments

---

11. ANALYTICS & PERFORMANCE DASHBOARDS

11.1 Features

- Subject averages, pass rates, term trends
- Identify top performers and at-risk students
- Predictive insights (e.g., declining performance)
- Correlate attendance with grades

11.2 Dashboards

- Class vs. Subject comparisons
- Historical trend visualization
- Performance alerts

---

12. ALERTS & NOTIFICATIONS

12.1 Triggers

- Low grades or failing subjects
- Missed or late assignments
- Upcoming exams or deadlines
- At-risk student warnings
- Parent notifications

---

13. PARENT & STUDENT PORTAL INTEGRATION

13.1 Features

- Secure access to report cards and grades
- Assignment submission and feedback
- Notifications for performance alerts and announcements
- Multi-child support

---

14. SECURITY & ROLE-BASED ACCESS

Role	Access Scope
Admin	Full access, setup, reports
Teacher	Assigned classes & students
Parent	Own child only
Student	Own grades, assignments, optional portal

---

15. MOBILE-FIRST DESIGN

15.1 Requirements

- Teacher-friendly mobile gradebook
- Mobile submission for assignments
- Parent/student mobile portals
- Offline data capture and sync

---

16. SCALABILITY

- Support thousands of students and multiple campuses
- Fast query and filtering for large datasets
- Extendable to new grading systems and curricula